**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 270 · BIOS, drivers and virtual machines

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does an OS driver do?

Q2. Define multi-user OS.

Q3. What executes before the OS is loaded?

Q4. Simplify (¬A∧B)∨(A∧B). Show factoring.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# I Do · Read new knowledge

### Key vocabulary

UEFI — Modern system boot firmware.

driver — Hardware-specific software interface.

VM — Virtual machine.

hypervisor — VM hosting software.

### New learning

Understand — BIOS/UEFI firmware initialises key hardware and starts the boot process.

Trace — Device drivers connect OS abstractions to the instructions and data expected by actual peripherals.

Evaluate — A hypervisor provides virtual machines for isolation, testing and consolidation, though CPU/RAM/storage overhead remains.

### Worked example

A printer is connected but software cannot print; name an OS dependency.

1. Predict first — A missing or incompatible driver can prevent correct communication with a device.

2. Model and check — Explain why each value follows from the earlier step.

# We Do · Practise with guidance

A teacher needs to test a new OS without replacing their existing install. Choose a solution.

```
Host OS + hypervisor + guest VM.
```

Predict first

A VM provides a reusable guest environment with snapshots, although it consumes host resources.

Model and check

Explain why each value follows from the earlier step.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

State roles of UEFI/BIOS, boot loader, driver and hypervisor. [4]

### Question 2

Explain how a compatible printer driver enables printing. [6]

### Question 3

Evaluate guest virtual machines for school testing of unfamiliar operating systems. [6]

## Exit Ticket · Answer before checking

A virtual machine has its own wholly separate physical processor.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Interfaces with specific hardware.

Q2. Supports separate user sessions and permissions.

Q3. Platform firmware and boot-loading code.

Q4. $B \wedge (\neg A \vee A) = B \wedge 1 = B$.

### You Do pages · Independent questions

Q1. Initialises platform [1], loads OS [1], accesses device [1], hosts/manages VMs [1].

Q2. App invokes OS service [1], OS queues request [1], driver translates to device instructions [2], incompatible interface may fail [2].

Q3. Isolation [2], snapshot/reset [1], multiple OSs [1], shared hardware/performance overhead [1], isolation/security not absolute [1].

### Exit Ticket · Model answer

False. It shares physical resources through a hypervisor.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**